

ENGR 102 Summer Bridge

Day 16 Student Quiz

Exam 2 Rehearsal: Loops, Lists, Mutation, and Filtering

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace skip, running state, collection, and early stop.

6 points

```
values = [3, -2, 5, 0]
result = []
running = 0

for index in range(len(values)):
    if values[index] < 0:
        continue
    running = running + values[index]
    if running > 6:
        result.append(running)
        break
    result.append(values[index] * 2)

print(result, running, index)
```

Give a complete reached-pass ledger, including skipped statements and the exact final output.

Question 2. Diagnose final-list evidence that disagrees with the total.

4 points

```
values = [4, -3, 7, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Give actual output, explain the mismatch, and write an indexed repair that makes the list and total describe the same final state.

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Question 3. Program construction

10 points

Write one complete Python program that satisfies every requirement.

- Read threshold as an integer and use values = [12, -4, 7, 20, 7].
- Loop by index and replace negative values with 0 before any summary.
- Append a final value to selected only when it is at least threshold and even.
- Compute the total of all final values and the index of the first largest final value without max or index.
- Print Final, Selected, Total, and Largest index on separate labeled lines. Give a threshold-equality test.

[illegible]

Test evidence / expected result